

Session_9

Q.1 Declare a 1D array called dailySteps with 7 elements to store your step count for each day of the week, assign sample values, and print each value using a loop.

A.1 `#include <stdio.h>`

```
int main() {
    int dailySteps[7] = {5000, 6200, 7500, 4800, 9000, 7000, 8500};

    int i;

    for (i = 0; i < 7; i++) {
        printf("Day %d: %d steps\n", i + 1, dailySteps[i]);
    }
    return 0;
}
```

Q.2 Create a 2D array called playlistRatings to store ratings for 3 Spotify playlists over 5 days (rows = playlists, columns = days). Fill it with sample numbers and print the ratings for the second playlist.

A.2 `#include <stdio.h>`

```
int main() {
    int playlistRatings[3][5] = {
        {5, 4, 5, 3, 4}, {4, 5, 3, 5, 5}, {3, 4, 4, 5, 4} };

    int i;

    printf("Ratings for the second playlist:\n");
    for (i = 0; i < 5; i++) { printf("Day %d: %d\n", i + 1, playlistRatings[1][i]);
    }
    return 0;
}
```

Q.3 Build a function that takes a 1D array of 7 integers representing your daily Zomato order amounts and calculates the average spend for the week. Hint: Use a loop to sum the values, then

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divide by the array length.

A.3 #include <stdio.h>

```
float calculateAverage(int orders[], int size) {
    int sum = 0;
    int i;

    for (i = 0; i < size; i++) {
        sum = sum + orders[i];
    }

    return (float)sum / size;
}

int main() {
    int dailyOrders[7] = {250, 180, 320, 150, 400, 275, 225};

    float average;
    average = calculateAverage(dailyOrders, 7);
    printf("Average weekly spend: %.2f\n", average);
    return 0;
}
```

Q. 4 Given a 2D array called cricketScores where each row represents an IPL match and columns represent runs scored by each team, write code to print the highest score from each match.

A.4

```
#include <stdio.h>

int main() {
    int cricketScores[4][2] = {{180, 165}, {145, 190}, {210, 205}, {175, 160}};
    int i;
    int highest;

    for (i = 0; i < 4; i++) {
```

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```
if (cricketScores[i][0] > cricketScores[i][1]) {  
    highest = cricketScores[i][0];  
}  
else {  
    highest = cricketScores[i][1];  
}  
printf("Match %d highest score: %d\n", i + 1, highest);  
}  
return 0;  
}
```
